

Partial Differential Equation (PDEs)

Tuesday, July 8, 2025 9:09 AM

1. Introduction to PDEs

1.1 PDEs are equations with partial derivative (as opposed to ordinary derivatives, integrals, etc).

1.2 Many "natural laws" are modelled as PDEs. Here are some famous ones

- wave equation: $\nabla^2 u - \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = 0$

- diffusion equation: $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \alpha \frac{\partial^2 u}{\partial x^2}$

- continuity equation: $\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = 0$

- heat equation: $\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$

- Euler-Lagrange equation: $\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0$

- Laplace's Equation: $\nabla^2 u = 0$ Poisson's Equation: $\nabla^2 u = f$

- Maxwell's Equations: $\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$ $\vec{\nabla} \cdot \vec{B} = 0$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

- Shrodinger Equation: $i\hbar \frac{\partial}{\partial t} \Psi(x,t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x,t) \right] \Psi(x,t)$

Other famous PDEs include: Helmholtz's equation, Euler's Equations, Navier-Stokes',

Other famous PDEs include: Helmholtz's equation, Euler's Equations, Navier-Stokes', Einstein field equations, Klein-Gordon equation, etc.

1.3 There are many advanced techniques for solving PDEs, but we will stick to simple ones such as:

- separation of variables
- Fourier Series
- Laplace Transforms

1.4: Deriving some PDEs from the Divergence Theorem

Continuity Equation for a compressible fluid

Suppose $\rho(x, y, z, t)$ is the density of a fluid and $\vec{v}(x, y, z, t)$ is the velocity. Then $\rho\vec{v}$ is the momentum, what does $\vec{\nabla} \cdot (\rho\vec{v})$ tell us?

For a region T , $\iiint_T \vec{\nabla} \cdot (\rho\vec{v}) dV$ is the total change in mass of fluid in that region.

But this is the same as: $\frac{d}{dt} \iiint_T \rho dV = \iiint_T \left(\frac{\partial \rho}{\partial t} \right) dV$ [Laplace's Rule]

This allows us to say: $\iiint_T \vec{\nabla} \cdot (\rho\vec{v}) dV = \iiint_T \left(\frac{\partial \rho}{\partial t} \right) dV$

Giving us: $\vec{\nabla} \cdot (\rho\vec{v}) = \frac{\partial \rho}{\partial t}$

EXAMPLE 2 Modeling of Heat Flow. Heat or Diffusion Equation

Physical experiments show that in a body, heat flows in the direction of decreasing temperature, and the rate of flow is proportional to the gradient of the temperature. This means that the velocity $\mathbf{v}$ of the heat flow in a body is of the form

$$(3) \quad \mathbf{v} = -K \text{ grad } U$$

where $U(x, y, z, t)$ is temperature, t is time, and K is called the *thermal conductivity* of the body; in ordinary physical circumstances K is a constant. Using this information, set up the mathematical model of heat flow, the so-called **heat equation** or **diffusion equation**.

Solution. Let T be a region in the body bounded by a surface S with outer unit normal vector $\mathbf{n}$ such that the divergence theorem applies. Then $\mathbf{v} \cdot \mathbf{n}$ is the component of $\mathbf{v}$ in the direction of $\mathbf{n}$, and the amount of heat leaving T per unit time is

$$\iint_S \mathbf{v} \cdot \mathbf{n} \, dA.$$

This expression is obtained similarly to the corresponding surface integral in the last example. Using

$$\operatorname{div}(\operatorname{grad} U) = \nabla^2 U = U_{xx} + U_{yy} + U_{zz}$$

(the Laplacian; see (3) in Sec. 9.8), we have by the divergence theorem and (3)

$$\begin{aligned} \iint_S \mathbf{v} \cdot \mathbf{n} \, dA &= -K \iiint_T \operatorname{div}(\operatorname{grad} U) \, dx \, dy \, dz \\ (4) \qquad \qquad \qquad &= -K \iiint_T \nabla^2 U \, dx \, dy \, dz. \end{aligned}$$

On the other hand, the total amount of heat H in T is

$$H = \iiint_T \sigma \rho U \, dx \, dy \, dz$$

where the constant σ is the specific heat of the material of the body and ρ is the density (= mass per unit volume) of the material. Hence the time rate of decrease of H is

$$-\frac{\partial H}{\partial t} = - \iiint_T \sigma \rho \frac{\partial U}{\partial t} \, dx \, dy \, dz$$

and this must be equal to the above amount of heat leaving T . From (4) we thus have

$$- \iiint_T \sigma \rho \frac{\partial U}{\partial t} \, dx \, dy \, dz = -K \iiint_T \nabla^2 U \, dx \, dy \, dz$$

or

$$\iiint_T \left(\sigma \rho \frac{\partial U}{\partial t} - K \nabla^2 U \right) dx \, dy \, dz = 0.$$

Since this holds for any region T in the body, the integrand (if continuous) must be zero everywhere; that is,

$$(5) \qquad \qquad \qquad \frac{\partial U}{\partial t} = c^2 \nabla^2 U \qquad \qquad c^2 = \frac{K}{\sigma \rho}$$